Afnan Mohammad Hafiz

ID : 23241004

Sec : [redacted]21

CSE [redacted]422

Ans. to the ques. no. 1

Given, the goal cell is at (4, 3).

$$h(P_1, P_2) = |x_1 - x_2| + |y_1 - y_2|$$

$$= |x_1 - 4| + |y_1 - 3|$$

$h(1,1) = |1-4| + |1-3| = 5$

$h(1,2) = |1-4| + |2-3| = 4$

$h(1,3) = |1-4| + |3-3| = 3$

$h(1,4) = |1-4| + |4-3| = 4$

$h(2,1) = |2-4| + |1-3| = 4$

$h(2,2) = |2-4| + |2-3| = 3$

$h(2,4) = |2-4| + |4-3| = 3$

$h(3,1) = |3-4| + |1-3| = 3$

$h(3,3) = |3-4| + |3-3| = 1$

$h(3,4) = |3-4| + |4-3| = 2$

$h(4,1) = |4-4| + |1-3| = 2$

$h(4,2) = |4-4| + |2-3| = 1$

$h(4,3) = 0$

~~Parent $(2,2)$~~

~~Child $(1,2),(2,1)$~~

	$(2,2)^{0+3}$
$(2,2)^{3}$	$(1,2)^{1+4}$ $(2,1)^{1+4}$
$(1,2)^{5}$	$(1,3)^{2+3}$ $(1,1)^{2+5}$ $(2,1)^{5}$
$(1,3)^{5}$	$(1,4)^{3+4}$ $(1,1)^{7}$ $(2,1)^{5}$
$(2,1)^{5}$	$(1,4)^{7}$ $(1,1)^{7}$ $(3,1)^{2+3}$
$(3,1)^{5}$	$(1,4)^{7}$ $(1,1)^{7}$ $(4,1)^{3+2}$
$(4,1)^{5}$	$(1,4)^{7}$ $(1,1)^{7}$
$(1,4)^{7}$	$(1,1)^{7}$ $(2,4)^{4+3}$
$(2,4)^{7}$	$(1,1)^{7}$ $(3,4)^{5+2}$
$(3,4)^{7}$	$(1,1)^{7}$ $(3,3)^{6+1}$
$(3,3)^{7}$	$(1,1)^{7}$ $(4,3)^{7+0}$
$(4,3)^{7}$	$(1,1)^{7}$

That is,

$$(4,3) \leftarrow (3,3) \leftarrow (3,4) \leftarrow (2,4) \leftarrow (1,4) \leftarrow (1,3) \leftarrow (1,2) \leftarrow (2,2)$$

Ans. to the ques. no. 2

a) $h_5(n) = 2h_2(n)$.

$h_9(n) = \max(h_2(n), h_3(n))$

Given, h_1 and h_2 is admissible but h_3 is not. For h_5, if the value of $h^*(n) = 5$ and $h(n) = 4$, then multiplying it by 2 will not align with the admissibility function -

$$h(n) \leq h^*(n)$$

Again, since h_3 is inadmissible, the max will pick h_3 as h_2 is admissible and therefore smaller than h_3.

b) h_7 dominates h_6 as it gives the maximum of both the function while h_6 gives average of both of the function. So, the values of h_6 is less than h_7.

c) h_7 is the best choice for A^* search because it goes deeper into the nodes as well as chooses the maximum among the two admissible heuristic.

Ans. to the ques. no. 3

a) True.

In hill-climbing, sideways moves allow the algorithm to traverse flat plateaus. It gives a chance to escape and potentially reach a global optimum from the plateau.

b) True

Simulated annealing makes a decision based only on the current and neighbouring states, not the entire search space.

c) True

Hill climbing is a local search algorithm. It only considers immediate neighbours and does not explore all states. So, it cannot explore the search space exhaustively.

d) False

Hill climbing makes greedy local decisions and can get stuck in local maxima, plateaus and ridges. So, it doesn't guarantee the shortest path in a maze.

Ans. to the ques. no. 4

a) $Y \xrightarrow{2} E \xrightarrow{4} T$

Actual distance is $2+4=6$.

Since $h_1 > h_2$, hence $h_1(Y)=6$

$h_2(Y)=5$

b) $\rightarrow \boxed{S^6}$

$S \; \boxed{B^8 \quad C^6 \quad D^9}$

$f(B) = g(B) + h(B) = 3+5=8$

$f(C) = g(C) + h(C) = 2+4=6$

$f(D) = g(D) + h(D) = 6+3=9$

C $\boxed{B^8 \quad F^8 \quad D^8}$

$f(F) = g(F) + h(F) = 6+2 = 8$

$f(D) = g(D) + h(D) = 5+3 = 8$

F $\boxed{B^8 \quad D^8 \quad T^8}$

$f(D) = g(D) + h(D) = 8+3 = 11$

$f(T) = g(T) + h(T) = 8+0 = 8$

T $\boxed{B^8 \quad D^8}$

That is,

$S \rightarrow C \rightarrow F \rightarrow T$

c) By using h_1, we get,

C $\boxed{S^8 \quad D^6 \quad F^6}$

$f(S) = g(S) + h(S) = 2+6 = 8$

$f(D) = g(D) + h(D) = 3+3 = 6$

$f(F) = g(F) + h(F) = 4+2 = 6$

D $\boxed{S^8 \quad F^6 \quad B^2 \quad T^7 \quad E^5}$

$f(B) = g(B) + h(B) = 7+5 = 12$

$f(T) = g(T) + h(T) = 7+0 = 7$

$f(E) = g(E) + h(E) = 5+0 = 5$

E $\boxed{S^8 \quad F^6 \quad B^{12} \quad T^7}$

That is,

$C \rightarrow D \rightarrow E$ which is 5. That is, its possible to get optimal path using h_1 or h_2.

d) $\underline{B \to E}$

$$h(B) \leq h(E) + C(B, E)$$

or, $5 \leq 4+3$

or, $5 \leq 7$ (consistent)

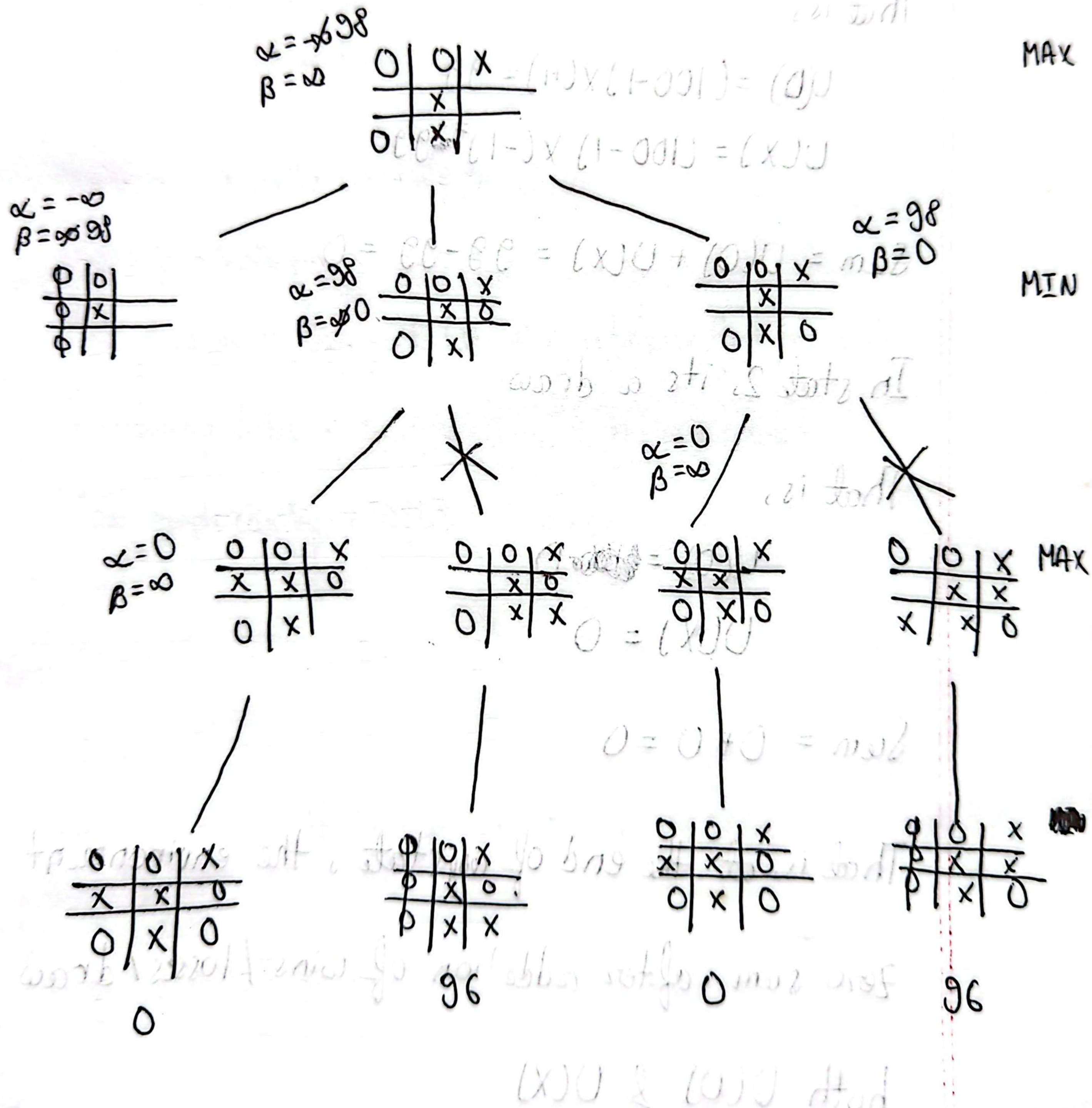
$\underline{B \to D}$

$$h(B) \leq h(D) + C(B, D)$$

or, $5 \leq 3+2$

or, $5 \leq 5$ (consistent)

Ans. to the ques. no. 5



b) In state 1, O wins

That is,

$U(O) = (100-1) \times (+1) = 99$

$U(X) = (100-1) \times (-1) = -99$

Sum $= U(O) + U(X) = 99 - 99 = 0$

In state 2, its a draw

That is,

$U(O) = 0$

$U(X) = 0$

Sum $= 0 + 0 = 0$

That is, at the end of any state, the environment is zero sum after addition of wins/losses/draw of both $U(O)$ & $U(X)$